



Grade 11 Geography

6. Factors Influencing Slope Development

Learning Guide

1. Introduction

- Slopes are **not static**.
- This means that they **change over time**.
- Their development is influenced by both:
 - **Natural factors**
 - **Human activities**

These factors affect:

- **Slope shape**
- **Slope stability**
- **Weathering**
- **Erosion**

★ Key idea: A slope changes continuously as material is weathered, eroded, moved, or removed. The diagram on page 3 shows how a **scarp retreats backwards** over time, while the **pediment becomes wider**, eventually contributing to a broader **pediplain**.

2. Climate

Climate strongly influences how quickly slopes weather, erode, and become unstable.

Rainfall

- Rainwater adds **extra weight** to slope material.
- A heavier slope has a greater chance of **collapse**.
- High rainfall also increases the **erosive power of rivers**.
- Rivers may erode the **base of a slope**, making it less stable.

★ Key idea: More water on a slope can increase both erosion and the risk of slope failure.

Meltwater

- Snow and glaciers can melt and produce large quantities of water.
- Meltwater has a similar effect to rainfall:
 - Adds water to slope material.
 - Increases erosion.
 - May increase slope instability.

Freeze-Thaw Weathering

Freeze-thaw occurs when water enters:

- Cracks
- Joints
- Rock pores

Process

1. Water enters cracks in the rock.
 2. The water freezes.
 3. Frozen water expands.
 4. Pressure is placed on the surrounding rock.
 5. The rock eventually breaks apart.
- Water freezing and expanding by about **9%**, placing stress on the rock until it breaks.
 - Broken material may then move downslope.

★ Sequence: Water enters crack → freezes → expands → rock breaks → material moves downslope

Avalanches

- Avalanches are common in areas that receive **heavy snowfall**.
- Snow and material move rapidly downslope.
- This can cause **rapid changes to slope shape**.

 **Climate summary:** Rainfall, meltwater, freezing, and snowfall can all speed up slope development.

3. Rock Type

The type of rock making up a slope affects how resistant the slope is to **weathering and erosion**.

Resistant Rock

- Resistant rocks weather and erode slowly.
- They generally create **more stable slopes**.
- Examples:
 - **Granite**
 - **Dolerite**

Soft / Less Resistant Rock

- Soft rocks weather and erode more easily.
- Slopes made from these rocks are:
 - Less stable.
 - More likely to experience movement.
 - More prone to landslides.

 **Key idea:** The more resistant the rock, the more stable the slope tends to be.

4. Soil Type

Soil characteristics also influence slope stability.

Thin Soil

- Thin soil provides less stable slope material.
- It can be more easily removed or moved.

Saturated Soil

- Soil becomes **saturated** when it contains a large amount of water.
- Water may come from:
 - Rainfall.
 - Irrigation.
- Saturated soil is **more prone to movement**.

Lack of Vegetation

- Without vegetation, there are no roots to hold soil firmly in place.
- This makes the soil more vulnerable to movement.

Possible Results

- **Mudslides**
- **Surface runoff**
- **Erosion**

 **Key idea:** Slope stability depends partly on the thickness of the soil and how much water it contains.

5. Vegetation

Vegetation can greatly increase slope stability.

How Dense Vegetation Stabilises Slopes

Roots Bind the Soil

- Roots hold soil particles together.
- This makes it more difficult for soil to move downslope.

Leaves Slow Surface Runoff

- Vegetation reduces how quickly rainwater reaches and flows across the soil.
- This reduces erosion.

 **Key idea:** Vegetation acts like a natural protective covering for slopes.

Effects of Removing Vegetation

A lack of vegetation can cause:

- **Faster erosion**
- Loose soil.
- Greater slope instability.
- Increased surface runoff.
- **Gully erosion**

Simple Relationship Vegetation removed → soil exposed → runoff increases → erosion increases → slope becomes unstable



6. Human Activities

People can speed up natural slope-development processes and make slopes less stable.



Deforestation

- Removes vegetation from slopes.
- This removes the roots that naturally stabilise soil.
- The slope becomes more vulnerable to:
 - Runoff.
 - Erosion.
 - Movement.



Key idea: Deforestation removes the slope's natural protection.



Road Construction

- Road construction often requires cutting into hillsides.
- This may:
 - Steepen the slope.
 - Remove some of its support.
 - Disturb the natural structure of the slope.



Mining and Quarrying

- Mining or quarrying near the **base of a slope** may weaken it.
- Removing material can reduce support for the slope above.
- This increases the risk of slope movement.



Overgrazing

- Too many animals remove vegetation.
- The soil becomes exposed.
- Soil may also become loosened.

Result

- Increased erosion.
- Greater instability.



Poor Ploughing

- Poor farming practices can loosen soil.
- Loose soil is more easily transported downslope by water or gravity.



Building on Steep Slopes

- Buildings can disturb the natural structure of the slope.
- They may contribute to slope instability.



Heavy Traffic

- Heavy vehicles create **vibrations**.
- These vibrations may increase the risk of slope movement.



Human activities summary: Deforestation + road construction + mining + overgrazing + poor farming + building + traffic = increased slope instability

7. How the Main Factors Affect Slope Development

Factor	Effect on the Slope
Climate	Adds water and weight, increases erosion and may trigger collapse
Rock type	Determines resistance to weathering and erosion
Soil	Stability depends on soil thickness and water content
Vegetation	Roots bind soil and reduce erosion
Human activities	May remove vegetation, weaken slopes or increase movement

8. How the Factors Work Together

Slope development is usually influenced by **more than one factor at the same time**.

For example:

Heavy rainfall → soil becomes saturated → slope becomes heavier → movement becomes more likely

Or:

Deforestation → roots removed → runoff increases → erosion increases → slope becomes unstable

Or:

Road construction → slope is cut and steepened → rainfall saturates exposed material → slope movement risk increases

 **Key idea:** Slope instability often results from several factors acting together.

9. Final Exam Summary

- Slopes **change over time**.
- Factors affecting slope development include:
 - **Climate**
 - **Rock type**
 - **Soil type**
 - **Vegetation**
 - **Human activities**
- Rainfall:
 - Adds weight.
 - Increases erosion.
 - May trigger collapse.
- Freeze-thaw weathering:
 - Water freezes.
 - Expands.
 - Breaks rock.
- Resistant rocks such as **granite and dolerite** generally form more stable slopes.
- Softer rocks weather and erode faster and are more prone to movement.
- Saturated soil is less stable.
- Vegetation:
 - Binds soil.
 - Reduces runoff.
 - Prevents erosion.
- Removing vegetation increases:
 - Runoff.
 - Gully erosion.
 - Instability.
- Human activities that may destabilise slopes include:
 - Deforestation.
 - Road construction.
 - Mining and quarrying.
 - Overgrazing.
 - Poor ploughing.
 - Building on steep slopes.
 - Heavy traffic.

 **Core idea:** Slope development is controlled by the interaction between climate, rock, soil, vegetation and human activity. When these factors reduce slope stability, erosion and mass movement become more likely

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